

DeepWaterSeg: High-Resolution Satellite Imagery Analysis for Water Body Extraction Using ResNet-U-Net

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Abstract—Accurate identification of water bodies is an important aspect of ecological oversight, disaster management, resource allocation, and conservation of aquatic life. To address this, the current research work *DeepWaterSeg* presents a deep learning technique for accurate identification of water bodies from high-resolution satellite images, including ponds, lakes, rivers, and more across diverse landscapes such as mountains, farmlands, and urban areas.

This research utilizes a ResNet50-based U-Net architecture trained on high-resolution satellite imagery, which enables precise mapping of water bodies. The model achieved impressive results with 96% pixel-wise precision and 92% accuracy, making it highly effective for segmentation tasks. The new *DeepWaterSeg* model works better than DMLU-Net and SDNet, which are both recent models. It gets 96% pixel-wise accuracy and 90% IoU on a variety of landscapes while being less computationally complex. This high level of accuracy makes it possible to reliably find both narrow and urban water bodies, showing that it could be useful for disaster response, resource management, and long-term ecosystem monitoring.

The proposed methodology can significantly aid in large-scale, real-time monitoring, contributing to improved disaster preparedness, drought assessment, and sustainable water resource utilization.

Index Terms—Water Body Segmentation, Satellite Imagery, Water Body Mapping, Disaster Management, Resource Allocation

I. INTRODUCTION

Water bodies such as lakes, rivers, ponds, and fishponds are vital components of the freshwater ecosystem. They support biodiversity, regulate climate, and provide crucial resources for agriculture, industry, and domestic needs. Proper upkeep and monitoring of these water bodies play a major role in maintaining ecological balance and are especially critical during disaster scenarios like floods. Therefore, identification, extraction, and mapping of water bodies are vital for enabling effective disaster response and resource allocation.

The table I mainly describes the freshwater percentage on Earth. In general, about 71 percent of the water on Earth

is found in oceans and seas, and the remaining 29 percent is freshwater; only a small fraction—around 1 percent—is actually accessible for human use [1]. at the same time, other water will drain into the oceans.

TABLE I
APPROXIMATE DISTRIBUTION OF FRESHWATER RESOURCES

Category	Approximate % of Freshwater
Inaccessible Freshwater	~99.697%
Accessible Freshwater	~0.3%
Human Use	~0.00026%
Wasted to Oceans	~0.0026%

According to [2], traditional methods of identifying water bodies during disasters can be quite slow, often taking a few days. However, with remote sensing, the same task can be done in just a couple of hours, making the response much faster and more efficient. Over the years, there has been a drop in aquatic biodiversity, making water body segmentation increasingly important for protecting these fragile ecosystems, where water body segmentation becomes crucial for conservation efforts.

Water segmentation faces multiple challenges such as distinguishing water bodies from complex terrains, variations in water color, and seasonal appearance changes.

To overcome these challenges, we propose a robust deep learning-based approach using a ResNet50 backbone with a U-Net architecture. Combined with effective preprocessing, our model achieves accurate segmentation even in complex, mixed-terrain regions. This solution supports near real-time mapping, reducing loss in disaster scenarios and enhancing freshwater ecosystem preservation. Recent deep learning models have performed well in water segmentation; however, many struggle to adapt to various terrains and necessitate substantial computational resources. This study addresses these challenges by presenting a lightweight ResNet-U-Net model trained on a heterogeneous dataset.

The model is made to find thin and irregular bodies of water accurately. It has better preprocessing steps that make edges sharper and cut down on false detections. Tests show that it works well and is easy to use while still being very accurate. The fig 1 mainly refers to the disaster response time [3] difference between the traditional sensing and by use of a remote sensing pipeline.

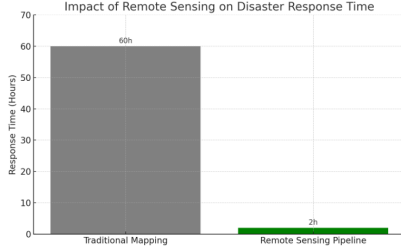


Fig. 1. Faster disaster response made possible through remote sensing

From this research, it is decided to opt resnet50-based U-Net architecture fed with 2481 high-quality satellite imagery and trained over 30 epochs. Literature review, Dataset, Model architecture, Mathematical notations, Model performance, and Conclusion and discussed later in the paper.

The fig 2 refers the research work which follows a traditional Pipeline where Data is collected from Kaggle [4], required preprocessing tasks are performed on the data, the model is trained, Multiple evolution metrics are considered to evaluate the model, performed testing against the model, Multiple plottings were used to analyze model performance, and lastly, the trained model is saved.

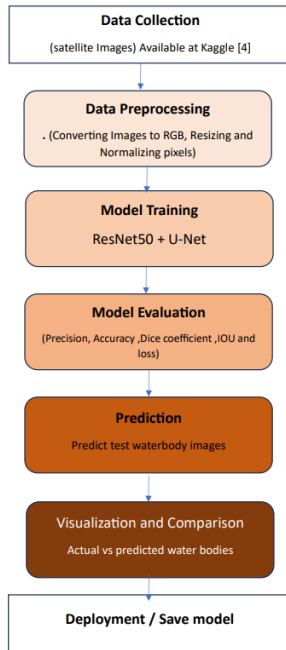


Fig. 2. Average Disaster Response Time

II. LITERATURE REVIEW

Accurate water body segmentation tasks are used to produce minimal results with traditional methods. However, in the recent past, due to new approaches, these tasks have been experiencing drastic improvements over the last couple of years, primarily due to advancements in Deep Learning, especially techniques associated with imagery data. Xu et al. [4] came up with DMLU-Net, which is specially trained for water body segmentation. This method uses dual encoders to improve accuracy and detect even thin edges. This approach enhances the water body extraction in varying resolution images. Chagas et al. [5] used Sentinel imagery for mapping and identification of small water bodies with the help of temporal data and spectral indices. This is not entirely preprocessing, but it provides valuable insights that can be integrated with neural networks. Ye et al. [6] introduced a high-precision water body reconstruction technique. This approach combines 3D stereo images with semantic segmentation, which enhances the accurate water body segmentation in high vegetative landscapes such as cities and urban areas. Attya et al. [7] used convolutional neural networks with an attention mechanism on satellite images. This model showed robust results with high-resolution images, precisely distinguishing between shadows, cloud reflections, etc. Jonnala et al. [8] came up with AER U-Net, an attention-based U-Net architecture with residual blocks and skip connections. It enriches the attention mechanism. Later on, the model's robustness is tested against Sentinel-2 imagery. Ni et al. [9] proposed SDNet (Sandwich Decoder Network), unlike the traditional U-Net, the proposed Sandwich Decoder Network replicates as a sandwich of multiple decode paths, which enhances the ability of capturing well-grained thin edges of water bodies from multiple landscapes Pech-May et al. [10] majorly focus on flood detection using Sentinel-1 SAR imagery and proposed that even with U-Net without any attention mechanism can attain optimal results while flooding. This study elevates the importance of SAR data Scarpetta et al. [11] have proposed a new and unique preprocessing approach for water body segmentation. This research work has proposed an NDWI dataset, and low-quality data samples are dropped to attain better results. At the end, the results have spiked up to 100 Ozdemir et al. [12] came up with a new proposal that consists of CLIP in a Visual Foundation Model (VFM) and a segment anything model, where trained on both satellite imagery and aerial images as well and attained a groundbreaking 94.41 Yuan et al. [13]. This research work is more into classifying real and fake water bodies in which involves 2 stages. where the 1st stage segments the water body, and at the 2nd stage classifies the water body as either a Real or a fake water body. This research work makes use of PoISAR imagery and leverages polarimetric features. Lakshminadh et al. [15], Rao et al. [16], and Reddy et al. [17] also provide foundational methodologies applicable to deep learning-based image analysis and classification. Hence, in this model, we have used ResNet50 as the backbone for the encoder and a U-Net model as the decoder, as [11] showed the better results

by using quality images, which improves the performance of water body segmentation. Earlier models such as DMLU-Net [4] and SDNet [9] have attained superior segmentation accuracy; however, they rely on intricate dual-encoder or multi-decoder configurations. This work, on the other hand, presents a simpler but still effective method by combining a single ResNet50 backbone with a U-Net decoder. This design keeps the accuracy high while lowering the computational load, which means it works well and consistently in different parts of the world.

III. METHODOLOGY

From fig 5 this model has been trained on high-resolution satellite imagery, as using low-resolution images has multiple setbacks that including improper identification of small and thin water bodies and imperfect boundary identification, which are neither ideal nor tolerable. The data set available at [14] consists of over 2481 high-resolution images and corresponding ground truth masked images as well for supervised learning. High-resolution imagery is perfect for identifying thin and narrow waterbodies such as curvy rivers, small ponds, and lakes. Due to the use of high-resolution images it enables the identification of urban water bodies, irrigation land water bodies, and wetland water bodies. High-resolution imagery enables the model to distinguish between water bodies and similar-looking shadows, which will eventually promote the model's accuracy. The data set not only includes a large number of images it also has have lot of diversity and variability of data. The data set consists of high-resolution imagery that captures water bodies from different landscapes and differently coloured water (blue colour fresh water, green colour algae, and leachy water), which enables the model to generalize more and can perform well on real-life imagery as well, which is both essential and commendable for practical deployment.

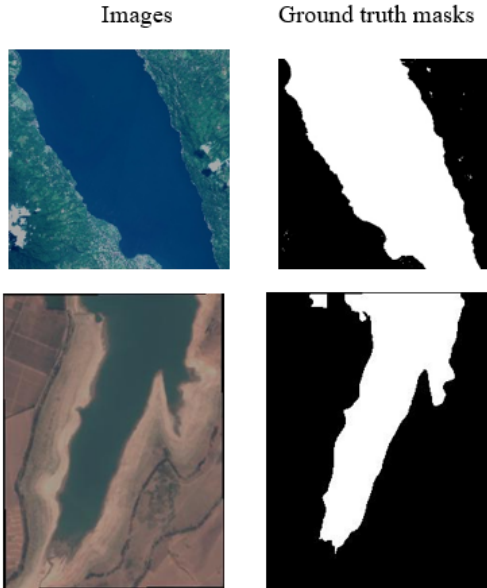


Fig. 3. Images obtained from the dataset and their ground truth masks

The figure 3 refers to the dataset, which consists of high-resolution satellite imagery of different sizes, as deep learning models expect it's inputs to have a fixed shape to run a filter on the images. The fig 4 describes the preprocessing steps that are performed before model training. Converting images to RGB from BGR this meets the expectations of the RESNET50 model as RESNET50 is pretrained on image data, we can use simple instructions with Computer Vision for RBG conversion. Converting masks to grayscale, in which the masked image will be mapped to binary values, which promotes the efficient water body extraction.

Resizing images and masks, as we are using RSENET50 with UNET architecture, which expects its input to have a fixed size for applying the filter, we must make sure that all images are of the same size and shape. Normalizing pixel values by dividing each pixel by 255, we can map our current image to a new space where each pixel ranges from 0 to 1.

Expanding mask dimensions: Grayscale masks are converted to 3D by adding a channel dimension, making them compatible with the model's output shape (height, width, 1). Here are few images before and after performing preprocessing tasks that include all the techniques mentioned above

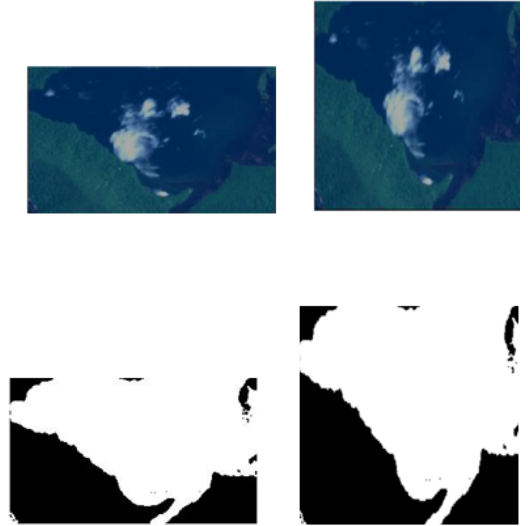


Fig. 4. Before vs After preprocessing

The ResNet50 U-Net model is designed with an Encoder and a Decoder. As ResNet50 is pretrained on Images, it can perform feature extraction precisely; hence, ResNet50 is used as an Encoder, and a U-Net-like architecture is used as a Decoder. Here is the model's high-level architecture. The ResNet50 encoder compresses the input image while keeping important features intact. The U-Net decoder then rebuilds the segmentation mask by slowly recovering the spatial details that were lost during compression. Input layer: The Input layer accepts images of dimensions $256 \times 256 \times 3$ where height=256 pixels, where height = 256 pixels, and has 3 colour channels (RGB). ResNet50 encoder consists of 5 stages, which are highly recommended for mapping an input feature from

higher spatial and semantic details to lower spatial or semantic details, which can be reconstructed using the U-Net Decoder. Bottleneck (Bridge) with the largest receptive field, encoding the most abstract and informative features of the input. These detailed features are subsequently sent to the decoder for progressive reconstruction of the segmentation mask.

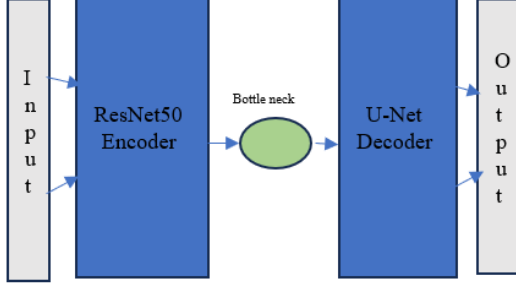


Fig. 5. Model architecture overview

U-Net Decoder in this model consists of 6 stages, where in each stage the input will be mapped into a new, higher-dimensional space. The output of the Encoder dimension is $8 \times 8 \times 2048$ is mapped to $256 \times 256 \times 1$. Here is a more detailed view of the U-Net decoder represented in fig 5 that includes upsamples and concatenation at each stage of the U-Net decoder.

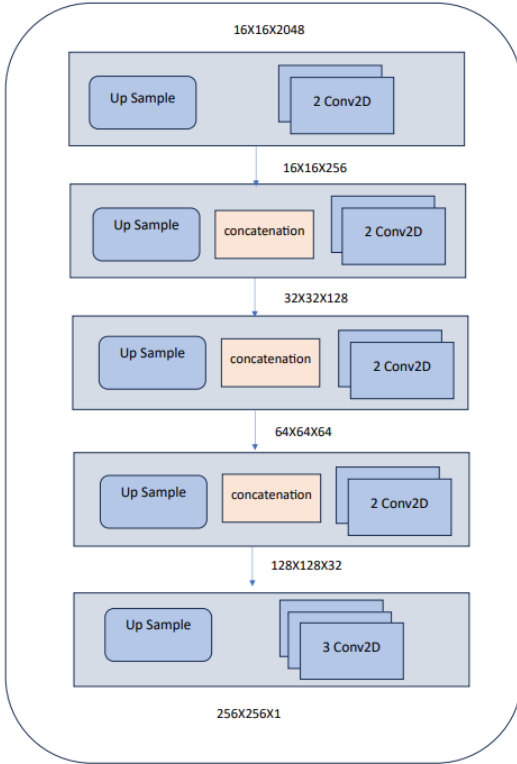


Fig. 6. Decoder Architecture

Decoder stage 1 ($8 \times 8 \rightarrow 16 \times 16$): Decoder begins with the encoder final output of dimension ($8 \times 8 \times 2048$) which has

referred in fig 6 with lavish semantic details (2048) but very low in Spatial data (8×8) hence in this layer the Spatial data is up sampled to 16×16 dimension and reduces the semantic detailing from 2048 to 256 consists of 3 layers including 1 concatenate and 2 Conv2D. Decoder Stage 2 ($16 \times 16 \rightarrow 32 \times 32$): The second stage of the decoder consists of 4 layers that include one upsample, one concatenate, and two Conv2D layers. This layer maps the input to higher spatial information (32×32) from 16×16 and reduces semantic information as well from 256 to 128 frames. Decoder Stage 3 ($32 \times 32 \rightarrow 64 \times 64$): The third stage of the U-Net encoder consists of four layers (one up sample, one concatenate, and two Conv2D). This stage maps the spatial information from 32×32 to 64×64 and reduces the semantic information from 128 frames to 64 frames.

Decoder Stage 4 ($64 \times 64 \rightarrow 128 \times 128$): Stage four of the decoder enriches the spatial information from 64×64 to 128×128 and reduces the semantic information to 32 frames from 64 frames. This stage consists of four layers same as stage 3 stage 2 and stage 2 (one up sample, one concatenate, and two conv2D layers). Decoder Stage 5 ($128 \times 128 \rightarrow 256 \times 256$): The last stage of U Net encoder – stage 5 consists of 4 layers unlike stage 4, 3 and 2 stage 5 consists of one up sample and 3 conv2D layers this layer maps the input spatial information to 256×256 from lower spatial information (128×128) and drastically reduces the semantic information from 32 to 1. The final dimension of the data is $256 \times 256 \times 1$.

TABLE II
TRAINING CONFIGURATION AND HYPERPARAMETERS

Parameter	Value
Input Shape	($256 \times 256 \times 3$)
Learning Rate	0.0001
Optimizer	Adam
Loss Function	Binary Cross Entropy
Epochs	30
Batch Size	8
Validation Split	0.2
Activation Functions	ReLU (hidden), Sigmoid (output)
Filters	$256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$
Smoothing Factor	0.000001

The table II mainly describes the parameters and their values present in the model. It mainly consists of parameters like shape, learning rate, batch size, loss, filters, e.t.c.

A. Mathematical Notation of ResNet50-U-Net Architecture

1) *Input*: Let N be the number of training images, each of spatial dimension 256×256 with 3 channels. The input is normalized between 0 and 1:

$$\mathbf{X} \in \mathbb{R}^{N \times 256 \times 256 \times 3} \quad (\text{normalized to } [0, 1])$$

Here, $\mathbf{X}$ is the input image tensor, representing N images, each with spatial resolution 256×256 and 3-channel RGB information.

2) *Encoder Layers (ResNet50)*: In ResNet50, the feature map at layer l is calculated as:

$$\mathbf{F}^l = \mathcal{H}^l(\mathbf{F}^{l-1}) + \mathbf{F}^{l-1}$$

Where:

- $\mathbf{F}^{l-1}$ is the previous layer's feature map.
- $\mathcal{H}^l$ is a function that applies convolution, batch normalization, and ReLU activation.
- The addition represents the residual connection (skip connection).

For example, at stage 3:

$$\mathbf{F}_3 = \mathcal{H}_3(\mathbf{F}_2) + \mathbf{F}_2$$

3) *Decoder Layers (U-Net-like)*: In the decoder, upsampling is performed to reconstruct the spatial resolution and reduce filter depth.

$$\begin{aligned} U(\mathbf{F}_l) &= \text{UpSampling2D}(\mathbf{F}_l) \\ \mathbf{C}_u &= \text{concatenate}(U(\mathbf{F}_l), \mathbf{S}_k) \\ \mathbf{F}_{l+1} &= \text{ReLU}(W * \mathbf{C}_u + b) \end{aligned}$$

Where:

- $U(\cdot)$ denotes the upsampling operation.
- $\mathbf{S}_k$ is the corresponding skip connection from the encoder.
- $*$ is the convolution operation.
- W and b are trainable parameters (weights and biases).

4) *Output Layer*: The final output $\hat{\mathbf{Y}}$ is produced using a sigmoid activation:

$$\hat{\mathbf{Y}} = \sigma(W_{\text{out}} * \mathbf{F}_{\text{last}} + b_{\text{out}}), \quad \text{where} \quad \sigma(x) = \frac{1}{1 + e^{-x}}$$

This layer outputs a probability map for binary segmentation using a 1×1 convolution followed by sigmoid activation.

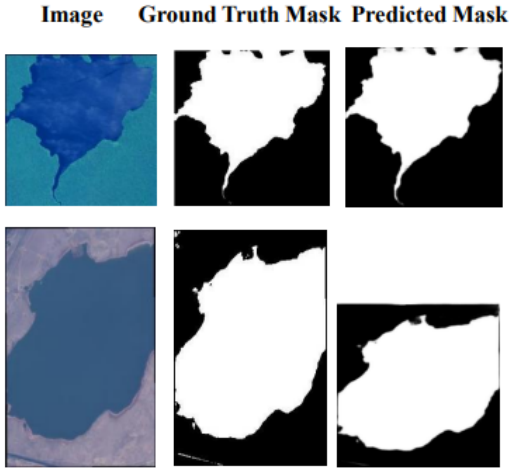


Fig. 7. Comparison of Ground Truth and Predicted Segmentation Masks for Water Body Detection

IV. RESULTS AND EVALUATION METRICS

The ResNet50 with U-Net architecture produced significant results in the water body segmentation task. The predicted segmentation masks closely resemble the ground truth masks. The figure 7 illustrates two examples comparing ground truth and predicted masks.

A. Quantitative Performance

After 30 training epochs, the model achieved exceptional results in several metrics, particularly in pixel-wise precision, which reached 96%. Other metrics also indicated strong performance:

- **Precision:** 96%
- **Dice Coefficient:** 91%
- **Accuracy:** 92%
- **Recall:** 86%
- **Intersection over Union (IoU):** 90%
- **Loss:** 6.2%

B. Loss Functions

The total loss function is a combination of Binary Cross Entropy (BCE) and Dice Loss with balancing factor λ :

$$L_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \cdot \log(\hat{y}_i) + (1 - y_i) \cdot \log(1 - \hat{y}_i)]$$

$$L_{\text{Dice}} = 1 - \frac{2 \sum_{i=1}^N y_i \hat{y}_i + \epsilon}{\sum_{i=1}^N y_i + \sum_{i=1}^N \hat{y}_i + \epsilon}$$

$$L_{\text{Total}} = (1 - \lambda) \cdot L_{\text{BCE}} + \lambda \cdot L_{\text{Dice}}$$

C. Performance Visualization

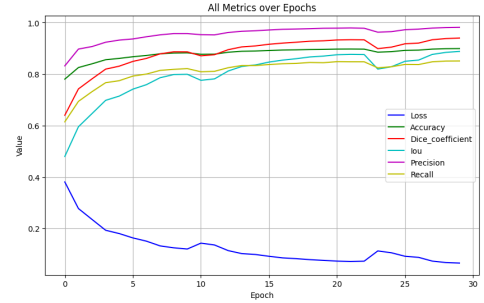


Fig. 8. Line plot of metrics (Precision, Accuracy, Dice, Recall, IoU, Loss) over 30 epochs.

The fig 8 mainly plots the metrics like precision, recall, Dice, IoU, and loss, which have been performed with over 30 epochs.

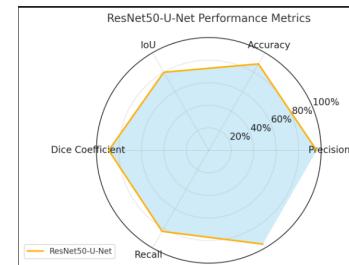


Fig. 9. Radar plot summarizing overall model performance.

The fig 9 shows the overall performance of the model on the basis of evaluation metrics.

D. Overall Performance Table

TABLE III
FINAL PERFORMANCE OF RESNET50-U-NET MODEL

Metric	Score
Pixel-wise Precision	96%
Accuracy	92%
Dice Coefficient	91%
Intersection over Union (IoU)	90%
Recall	86%
Loss	6.2%

The table III refers to the final performance of the model and the score that had obtained with metrics like precision, Accuracy, Dice coefficient, IoU, Recall, and loss.

E. Comparison with Other Works

Recent models such as DMLU-Net [4] and SD-Net [9] have demonstrated notable performance in segmentation tasks, especially on thin boundaries and urban terrains. However, they come with higher computational complexity and training time. Our ResNet50-U-Net model achieves better results with

TABLE IV
COMPARISON OF DEEPWATERSEG WITH EXISTING MODELS

Model	Precision (%)	IoU (%)
DMLU-Net [4]	91	86
SDNet [9]	92	88
DeepWaterSeg (proposed)	96	90

a lightweight architecture:

- Outperforms models like those in [5], [8] which reported 91% precision.
- Attains 96% precision, 91% Dice coefficient, and 90% IoU with less architectural complexity.
- Trained on a diverse dataset including rivers, ponds, and lakes—enabling better generalization.

In contrast to works such as [6], [7], [13], which are trained on specific terrains, our model generalizes well across various natural water bodies and edge conditions. This is crucial for real-world applications like small water body identification and land-water boundary segmentation [12].

V. CONCLUSION

In this research, we successfully employed deep learning techniques to develop a ResNet50-U-Net architecture for the accurate extraction of water bodies from high-resolution satellite imagery. The model demonstrated robust performance across diverse water bodies, including rivers, lakes, and ponds, in terrains with varying geographical features.

This study introduces several technical advancements that have practical implications beyond segmentation accuracy. The proposed method significantly enhances capabilities in key domains such as:

- **Disaster Response:** Faster and automated mapping of flood-affected areas can reduce average disaster response time.

- **Sustainable Development:** Accurate water mapping supports better planning and monitoring of freshwater resources.
- **Resource Management:** Helps in efficient allocation and monitoring of freshwater for agriculture, consumption, and industry.
- **Freshwater Preservation:** Supports the conservation of freshwater ecosystems by monitoring changes in water bodies over time.
- **Habitat Protection:** Enables tracking of habitat degradation and promotes the preservation of aquatic biodiversity.

The integration of deep learning into remote sensing tasks not only automates the process of water body extraction but also contributes directly to achieving several United Nations Sustainable Development Goals (SDGs). The proposed approach thus fosters environmental accountability, facilitates informed decision-making, and offers a scalable solution for global freshwater management challenges.

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